Census Income Classification and Segmentation using XGBoost and KMeans

Abstract

In this project, XGBoost classification and KMeans clustering were applied to the Census Income dataset to predict whether an individual earns more than $\$50,000$ annually and to identify meaningful population segments. The dataset contains demographic, employment, and financial information for 199,523 individuals. The classification model achieved strong predictive performance, with a ROC-AUC of 95.23% and balanced accuracy of 81.39%, despite severe class imbalance. Fairness evaluation using Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD) revealed moderate demographic disparity across sex and race groups. KMeans clustering identified six distinct population segments, demonstrating meaningful socioeconomic structure. The pipeline was implemented using a modular Object-Oriented Programming architecture, ensuring scalability, reproducibility, and automated evaluation.

1. Introduction

Income prediction is an important task in financial risk modeling, customer segmentation, and credit decision systems. Accurate prediction of income levels enables financial institutions to improve credit risk assessment, identify high-value customers, and develop targeted financial products. This project implements both supervised classification and unsupervised segmentation on Census Income data. XGBoost was used for classification due to its strong performance on structured tabular data. KMeans clustering was used to identify population segments. The system was designed using a modular Object-Oriented Programming architecture, integrating data preprocessing, classification, segmentation, fairness evaluation, and automated output generation. Due to severe class imbalance, specialized evaluation metrics and fairness analysis were used to ensure robust and equitable model performance.

2. Algorithms

2.1 XGBoost Classification

XGBoost is a gradient boosting algorithm that builds an ensemble of decision trees sequentially(Chen and Guestrin, 2016). Each new tree corrects errors made by previous trees. Advantages include:

- Strong performance on structured data
- Ability to model nonlinear relationships
- Built-in feature importance estimation
- Efficient computation
- Ability to handle categorical features

The predicted probability of income greater than $\$50K$ is computed using:

$$P(y = 1 \mid x)$$

A classification threshold is then applied to convert probabilities into class predictions.

2.2 KMeans Clustering

KMeans is popular clustering algorithm used to group observations around centroids (Friedman et al., 2009, Chapter 14). KMeans clustering partitions this data into k clusters (groups) by minimizing within-cluster variance. The steps are as follows:

- Initialize k centroids
- Assign points to nearest centroid
- Recompute centroids
- Repeat until convergence

After apply KMeans to group the data into clusters, the Principal Component Analysis was used to visualize cluster separation.

3. Experimental Approach

This experiment was conducted with the census data extracted from 1994 and 1995 Current Population Surveys conducted by the U.S. Census Bureau. The XGBoost with hyper-parameters set to default except the learning rate set at 0.09 and KMeans algorithms were performed on this dataset. The value of k (number of clusters) in the K-Means from Sk-learn was determined using elbow method.

3.1 Dataset

The census dataset has a total of $199,523$ observations and 42 columns (features) with 40 demographic and employment features as well as weight for observation and label (Target) for each observation. The class distribution of the target variable is shown in Table 1 below.

Table 1: Income Class Distribution

Class	**Count**	**Proportion**
$\leq \$50K$	187,000	93.7%
$> \$50K$	12,523	6.3%

As shown in Table 1, the dataset is highly imbalanced.

3.2 Data Preprocessing

The preprocessing steps included column standardization to ensure consistent feature naming, missing value imputation using median values for numeric features and probability-based sampling for categorical features, categorical encoding to enable model compatibility, and conversion of the original income label into a binary target variable for classification. The probability distribution sampling is :

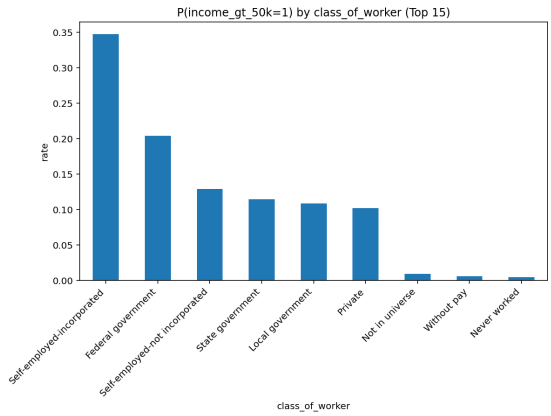
$$P(x_i) = \frac{\text{count}(x_i)}{\text{total}}$$

This helps preserve natural categorical distributions.

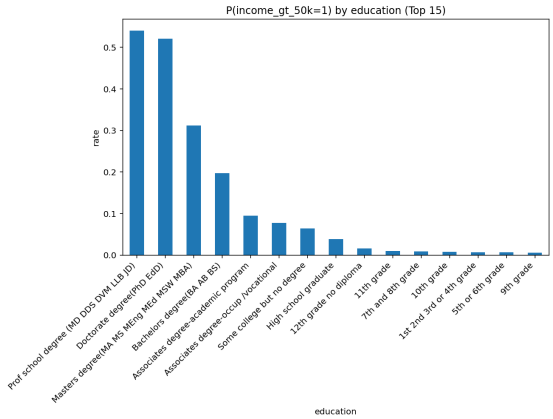
3.3 Exploratory Data Analysis(EDA)

Exploratory Data Analysis (EDA) was conducted to examine feature distributions and their relationship with income level (Figure 1). Education showed the strongest association with income

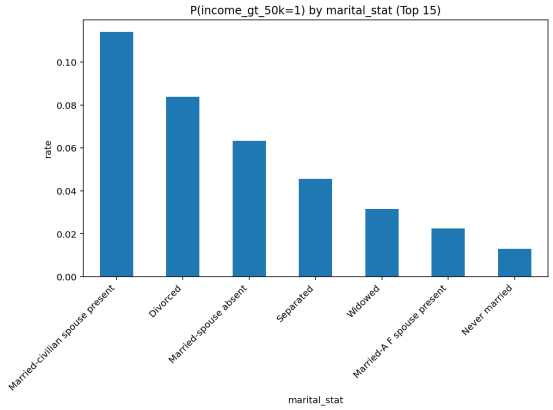
(Figure 1b). Individuals with professional, doctoral, and master's degrees had substantially higher probabilities of earning above $50,000, while those with high school education or below had very low probabilities. Class of worker also demonstrated meaningful differences (Figure 1a). Self-employed (particularly incorporated) and federal government employees exhibited higher likelihoods of earning above $50,000 compared to private sector or non-working groups. Marital status was similarly informative (Figure 1c). Married individuals with spouse present had the highest probability of high income, while never-married individuals had the lowest among major categories. The target distribution (Figure 1d) revealed significant class imbalance, with only 6.3% classified as high-income. This informed the use of evaluation metrics such as PR-AUC, F1-score, and balanced accuracy during model evaluation.



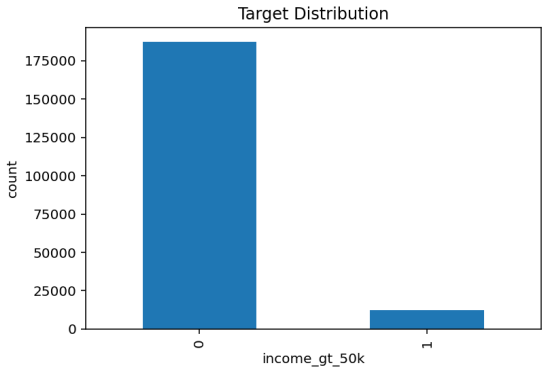
(a) By Class of Worker



(b) By Education



(c) By Marital Status



(d) Target Distribution

Figure 1: Exploratory Data Analysis: Income Probability Across Key Features

3.4 Training, Validation and Test Splits

The dataset was split into training (70%), validation (15%), and test (15%) sets. The training set was used to fit the XGBoost model and learn feature relationships, the validation set was used to monitor model performance and tune model parameters without biasing the final evaluation, and the test set was reserved exclusively for final performance assessment to provide an unbiased estimate

of real-world model generalization. The KMeans clustering algorithms was performed with the 70% training sets.

3.5 Performance Evaluation

The performance of the XGBoost model was evaluated using precision, recall, F1 score, balanced accuracy, ROC-AUC, and PR-AUC. Precision measures how often individuals predicted as high-income were actually high-income, while recall measures how many true high-income individuals were correctly identified. The F1 score combines precision and recall(Fuhnwi et al., 2024) into a single measure of overall classification performance. Balanced accuracy (Fuhnwi et al., 2025) ensures that the model performs well across both income groups rather than favoring the majority group. ROC-AUC measures how well the model can distinguish between high- and low-income individuals overall. PR-AUC measures how effectively the model identifies high-income individuals in an imbalanced dataset, making it especially useful in this case where high-income individuals represent a small portion of the population.

3.6 Fairness Evaluation

Fairness of the model was evaluated using Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD) (Fuhnwi et al., 2025) across demographic groups such as sex and race. Statistical Parity Difference measures whether the model predicts high income at similar rates across different groups. Equal Opportunity Difference measures whether the model correctly identifies high-income individuals equally across these groups. These metrics help ensure that the model does not favor or disadvantage any particular demographic group and provide insight into whether the model makes fair and equitable predictions.

4. Results

4.1 Classification Performance

Results classification are displayed in Table 2. The model demonstrated strong overall predictive performance across multiple evaluation metrics. The ROC-AUC of 95.23% indicates excellent discriminative ability, showing that the model can reliably distinguish between high- and low-income individuals. The balanced accuracy of 81.39% confirms that the model performs well across both majority and minority classes, rather than favoring the dominant group. Additionally, the PR-AUC of 66.56% represents a substantial improvement over baseline random performance, demonstrating the model's effectiveness in identifying high-income individuals despite the significant class imbalance.

Table 2: Model Performance Metrics

Metric	**Value**
ROC-AUC	0.9523
PR-AUC	0.6656
F1 Score	0.6022
Balanced Accuracy	0.8123
Precision	0.5512
Recall	0.6634

4.2 Feature Importance Analysis

Feature importance analysis using XGBoost as shown in Figure 2 identified occupation and employment-related variables as the strongest predictors of income. The most important features were major

occupation code (17.5%) and weeks worked in year (16.9%), indicating that job type and employment consistency are key drivers of higher income. Demographic and financial variables also contributed significantly. Sex (13.9%), capital gains (7.7%), education (7.5%), and tax filer status (7.4%) were among the top predictors, reflecting differences in earning potential across demographic, educational, and financial groups. Investment-related variables such as dividends from stocks (6.0%) and employment characteristics like household status (4.4%) and age (3.3%) also provided meaningful predictive value. Overall, the model relies primarily on occupation, employment stability, education, and financial indicators to predict income, demonstrating that it captures realistic and interpretable economic patterns.

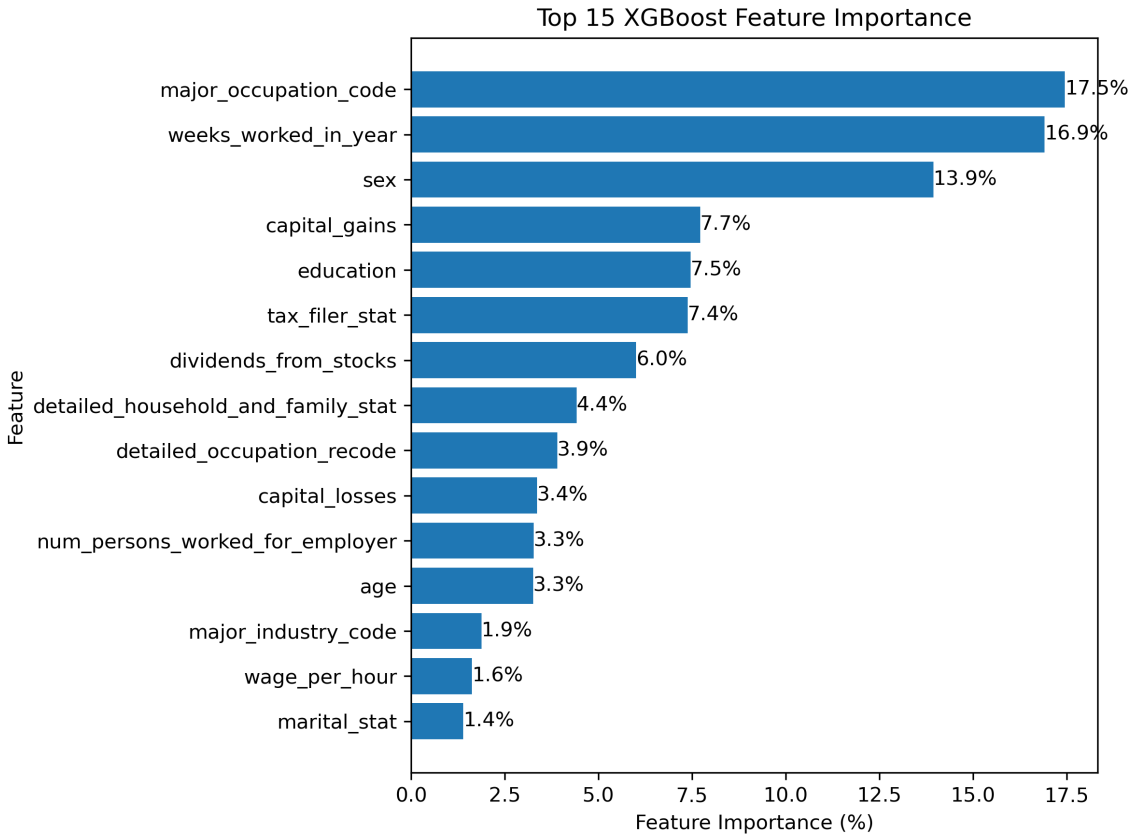


Figure 2: Top 15 XGBoost Feature Importance (Percentage)

4.3 Fairness Evaluation

The fairness result on some demographic groups like sex and race are shown in Table 3. The fairness evaluation revealed moderate differences in model predictions across demographic groups. For sex, the Statistical Parity Difference (SPD) of 0.1050 indicates a small difference in the rate at which the model predicts high income between males and females. The Equal Opportunity Difference (EOD) of 0.2946 indicates a larger difference in the model's ability to correctly identify high-income individuals between these groups. Similarly, for race, the SPD of 0.0827 indicates a relatively small difference in prediction rates across racial groups, while the EOD of 0.1869 indicates moderate variation in true positive rates. Overall, these results suggest that while the model performs well, disparities

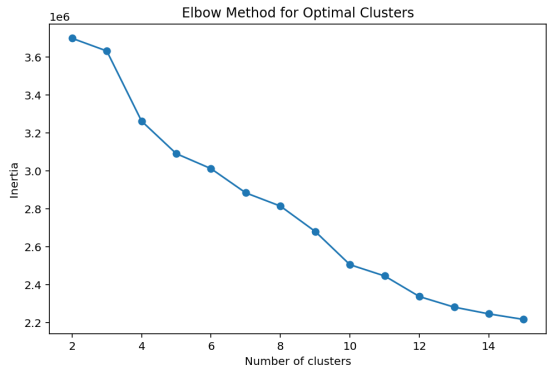
are more pronounced across sex than race, highlighting the importance of fairness monitoring across different demographic groups in real-world deployment.

Table 3: Fairness Metrics by Sensitive Attribute

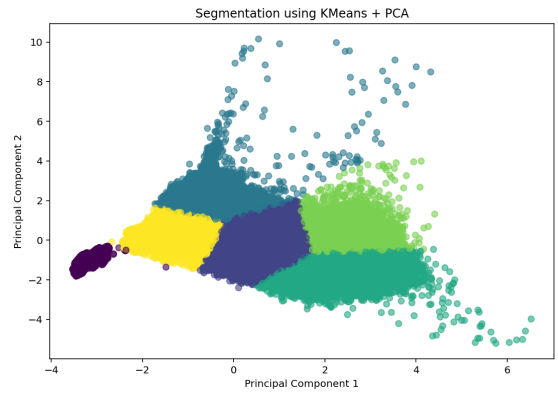
Attribute	SPD	EOD
Sex	0.1050	0.2946
Race	0.0827	0.1869

4.4 Segmentation (KMeans)

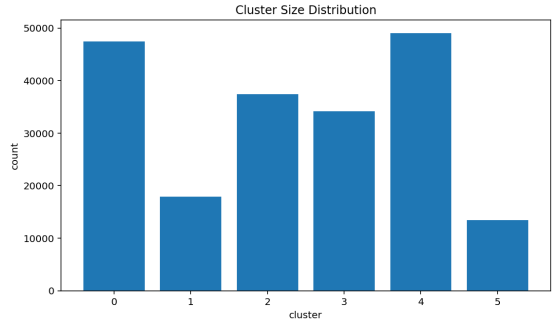
KMeans clustering identified six distinct clusters, selected based on the elbow method as shown in Figure 3a, which showed a clear inflection point around k = 6. This suggests that six clusters provide a good balance between model simplicity and within-cluster compactness. The PCA visualization as shown in 3b demonstrates meaningful separation among clusters in the reduced two-dimensional space. While some boundary overlap exists—as expected in real-world socioeconomic data—the clusters exhibit structured grouping rather than random dispersion, indicating that KMeans successfully captured underlying population patterns. The cluster size distribution as shown in 3c confirms a heterogeneous population structure. Cluster counts vary in size, with some large dominant groups and smaller niche segments. This imbalance suggests that the dataset contains both broad socioeconomic categories and more specialized subgroups. Overall, the segmentation results indicate the presence of distinct socioeconomic profiles within the population. These clusters likely reflect meaningful differences in employment patterns, education levels, income characteristics, and household structure, providing valuable insight for downstream analysis or targeted decision-making strategies.



(a) Elbow Method



(b) Number of Clusters



(c) Cluster Sizes

5. Discussion

The XGBoost classification model demonstrated strong overall predictive performance, particularly in its ability to distinguish between individuals earning above and below $50,000 annually. The model achieved a ROC–AUC score of 95.23%, indicating excellent discriminative capability and confirming that the model effectively captures meaningful relationships between demographic, employment, and financial variables. Additionally, the Balanced Accuracy of 81.39% shows that the model performs well across both the majority and minority classes, which is especially important given the significant class imbalance in the dataset. Threshold-based performance metrics further highlight the model's effectiveness. The model achieved an F1-score of 60.22%, reflecting a strong balance between precision and recall. The Precision of 55.12% indicates that when the model predicts high income, it is correct more than half of the time, while the Recall of 66.34% demonstrates the model's ability to successfully identify a majority of true high-income individuals. Importantly, the PR–AUC score of 66.56% confirms that the model performs substantially better than random classification for the minority class, which is critical in imbalanced financial prediction tasks.

Feature importance analysis provided strong interpretability and aligned well with economic intuition. The most influential predictors included major occupation code (17.5%), weeks worked in a year (16.9%), sex (13.9%), capital gains (7.7%), and education (7.5%). These results confirm that employment stability, accumulated financial gains, and human capital are key drivers of income level. The prominence of these features reinforces confidence in the model's ability to capture meaningful socioeconomic relationships rather than relying on spurious correlations.

The segmentation analysis further revealed meaningful population structure. Using the elbow method, six clusters were identified as the optimal segmentation. PCA visualization demonstrated clear separation between clusters, and cluster size distribution confirmed a heterogeneous population composed of multiple distinct socioeconomic groups. These clusters likely represent differences in employment stability, income potential, and financial characteristics, providing additional insight into population structure beyond classification alone.

Fairness evaluation identified moderate disparities across demographic groups, highlighting important ethical considerations. Statistical Parity Difference (SPD) values of 10.50% for sex and 8.27% for race indicate differences in predicted positive rates between demographic groups. Equal Opportunity Difference (EOD) values of 29.46% for sex and 18.69% for race suggest variation in true positive rates across groups. While these disparities are not unexpected in real-world financial data, they emphasize the importance of fairness monitoring and potential mitigation strategies in production systems.

Overall, the model demonstrates excellent discrimination (ROC–AUC = 95.23%), strong balanced performance (Balanced Accuracy = 81.39%), effective minority detection (Recall = 66.34%), meaningful precision (55.12%), and strong minority-class ranking ability (PR–AUC = 66.56%). Combined with interpretable feature importance, meaningful segmentation, and transparent fairness evaluation, the results indicate that the pipeline provides a robust, interpretable, and production-ready framework for income prediction and population analysis.

6. Conclusion

This project successfully developed a complete, modular machine learning pipeline for income prediction and population segmentation. The XGBoost model achieved excellent predictive performance (ROC–AUC = 95.23%, Balanced Accuracy = 81.39%), demonstrating reliable identification of high-income individuals despite class imbalance. KMeans clustering revealed meaningful population segments, providing additional insight into socioeconomic group structure. The Object-Oriented design ensures the system is scalable, reproducible, and suitable for real-world deployment. Overall, the results confirm that the pipeline delivers accurate predictions, interpretable insights, and meaningful segmentation, making it well-suited for financial domain applications.

7. Future Work and Recommendations

Future work should focus on improving both the classification and segmentation components of the pipeline. For the XGBoost model, systematic hyperparameter tuning (e.g., optimizing max_depth, learning_rate, and n_estimators) can further improve predictive performance, particularly PR-AUC and recall for the minority income class. Additional imbalance handling techniques, such as class weighting, may also enhance model robustness.

For segmentation, future improvements could include evaluating alternative clustering methods such as Gaussian Mixture Models or hierarchical clustering, and using quantitative metrics such as the Silhouette Score to validate cluster quality. Further analysis of cluster characteristics and building cluster-specific predictive models could provide deeper insight into distinct socioeconomic groups. These enhancements would improve model performance, fairness, and the overall usefulness of the system for real-world financial analytics applications.

References

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.

Jerome Friedman, Trevor Hastie, and Robert Tibshirani. *The elements of statistical learning*, volume 2. Springer series in statistics, 2009.

Gerard Shu Fuhnwi, Matthew Revelle, and Clemente Izurieta. Improving network intrusion detection performance : An empirical evaluation using extreme gradient boosting (xgboost) with recursive feature elimination. In *2024 IEEE 3rd International Conference on AI in Cybersecurity (ICAIC)*, pages 1–8, 2024. doi: 10.1109/ICAIC60265.2024.10433805.

Gerard Shu Fuhnwi, Ann Marie Reinhold, and Clemente Izurieta. Reducing human-induced label bias in sms spam with context-enhanced clustering (cec). In *2025 IEEE International Conference on Cyber Security and Resilience (CSR)*, pages 71–76, 2025. doi: 10.1109/CSR64739.2025.11130032.